Due Date

You must *submit* the source code for the solution to this lab exercise to *Moodle* by

Monday, December 1, 2025

in order to receive full credit for this work. You must also *demonstrate* the solution to the instructor during class, at the earliest opportunity.

Compute Greatest Common Divisor using Recursion

In class we discussed a "brute force" method of computing Greatest Common Divisor (using a loop), as well as a **recursive** algorithm, known as ***Euclid's Algorithm***, which is stated below:

If **m** and **n** are positive integers, **gcd(m, n)** can be defined as follows:

- If **m%n** is **0**, then **gcd(m,n)** is **n**.
- Otherwise, **gcd(m,n)** is **gcd(n, m%n)**.

The requirement for this Lab exercise is to **write a recursive method** to find the GCD, using *Euclid's Algorithm*. Write a test program that accepts two integer inputs and displays their GCD. Include an interactive outer loop that allows the user to repeat the process for other numbers.

Instrumenting the Code

We have seen before that adding print statements at key locations in a program can help illustrate exactly how that program works.

At the entry point of the **`gcd`** method, add a statement to output the calling parameters **`m`** and **`n`**. This will allow you to see how many times the recursive method was called, and what the calling parameters were each time the method was called.

Before each return from the **`gcd`** method, add a statement to output the returned value, as well as the input parameters that were used for that particular invocation of **`gcd`**. (Also indicate whether this return is from a "base case" or a "recursive case".)

Sample Output

The examples that follow show correct output for several test cases. (In these examples, the text that the user types is shown in **BOLD** font. The actual input / output will all be displayed in the same font.)

Sample Output

```
Enter two integers (or 'q' to exit): 511 22
Entering 'gcd' method:  m = 511, n = 22
Entering 'gcd' method:  m = 22, n = 5
Entering 'gcd' method:  m = 5, n = 2
Entering 'gcd' method:  m = 2, n = 1
Returning 'gcd' value = 1 (BASE CASE:      m=2, n=1)
Returning 'gcd' value = 1 (recursive case: m=5, n=2)
Returning 'gcd' value = 1 (recursive case: m=22, n=5)
Returning 'gcd' value = 1 (recursive case: m=511, n=22)
The GCD of 511 and 22 is 1.
```

Sample Output

```
Enter two integers (or 'q' to exit): 9938 8
Entering 'gcd' method:  m = 9938, n = 8
Entering 'gcd' method:  m = 8, n = 2
Returning 'gcd' value = 2 (BASE CASE:      m=8, n=2)
Returning 'gcd' value = 2 (recursive case: m=9938, n=8)
The GCD of 9938 and 8 is 2.

Enter two integers (or 'q' to exit): 37 1023
Entering 'gcd' method:  m = 37, n = 1023
Entering 'gcd' method:  m = 1023, n = 37
Entering 'gcd' method:  m = 37, n = 24
Entering 'gcd' method:  m = 24, n = 13
Entering 'gcd' method:  m = 13, n = 11
Entering 'gcd' method:  m = 11, n = 2
Entering 'gcd' method:  m = 2, n = 1
Returning 'gcd' value = 1 (BASE CASE:      m=2, n=1)
Returning 'gcd' value = 1 (recursive case: m=11, n=2)
Returning 'gcd' value = 1 (recursive case: m=13, n=11)
Returning 'gcd' value = 1 (recursive case: m=24, n=13)
Returning 'gcd' value = 1 (recursive case: m=37, n=24)
Returning 'gcd' value = 1 (recursive case: m=1023, n=37)
Returning 'gcd' value = 1 (recursive case: m=37, n=1023)
The GCD of 37 and 1023 is 1.

Enter two integers (or 'q' to exit): 5323 77
Entering 'gcd' method:  m = 5323, n = 77
Entering 'gcd' method:  m = 77, n = 10
Entering 'gcd' method:  m = 10, n = 7
Entering 'gcd' method:  m = 7, n = 3
Entering 'gcd' method:  m = 3, n = 1
Returning 'gcd' value = 1 (BASE CASE:      m=3, n=1)
Returning 'gcd' value = 1 (recursive case: m=7, n=3)
Returning 'gcd' value = 1 (recursive case: m=10, n=7)
Returning 'gcd' value = 1 (recursive case: m=77, n=10)
Returning 'gcd' value = 1 (recursive case: m=5323, n=77)
The GCD of 5323 and 77 is 1.

Enter two integers (or 'q' to exit): 5328 68
Entering 'gcd' method:  m = 5328, n = 68
Entering 'gcd' method:  m = 68, n = 24
Entering 'gcd' method:  m = 24, n = 20
Entering 'gcd' method:  m = 20, n = 4
Returning 'gcd' value = 4 (BASE CASE:      m=20, n=4)
Returning 'gcd' value = 4 (recursive case: m=24, n=20)
Returning 'gcd' value = 4 (recursive case: m=68, n=24)
Returning 'gcd' value = 4 (recursive case: m=5328, n=68)
The GCD of 5328 and 68 is 4.

Enter two integers (or 'q' to exit): q
```